



Modeling liquid rocket engine coolant flow and heat transfer in high roughness channels

Beatrice Latini^{a,b}, Matteo Fiore^{a,b}, Francesco Nasuti^{a,b,*}

^a Sapienza University of Rome, Italy

^b Dipartimento di Ingegneria Meccanica e Aerospaziale, Via Eudossiana 18, Rome I-00184, Italy

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ABSTRACT

The design of thrust chambers of modern liquid rocket engines is moving towards the use of additive layer manufacturing techniques, which are replacing traditional techniques because of its inherent advantages of customization and flexibility of design and prototyping phase that in the end leads to advantages in terms of production costs and times. In this framework, the realization of suitable cooling channels, which are characterized by tiny cross section, can lead to high relative surface roughness, as a result of the manufacturing process. Despite the increased friction loss, high roughness channels are considered often with interest because of the corresponding increase of heat transfer and thus cooling efficiency. However, the correlation between the increase of friction and heat transfer changes at high roughness. As a consequence, the commonly used assumption for heat transfer prediction at low roughness may lead to order of magnitude errors if extended to high roughness channels. To respond to the request of a reliable heat transfer prediction model, the correction introduced by Aupoix to the Spalart-Allmaras turbulence model to take into account for high roughness in external flows is extended and validated here for the application to high roughness channels. The implementation of the proposed correction allows to predict realistic values of the convective heat transfer coefficient, in agreement with the correlations reported in the literature.

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1. Introduction

In the design, realization and testing of liquid propellant rocket engines (LRE), the huge amount of power released in a relatively small volume makes cooling of thrust chamber walls not only necessary but also one of the most challenging issues. Among the different cooling techniques, the regenerative cooling system is the most employed because of its high efficiency. This technique uses one of the propellants on board which flows through small channels surrounding the thrust chamber to cool the wall and keep it within the allowable range of temperatures that guarantee its resistance to the expected range of static, dynamic and fatigue stresses. Additionally, in special cases [1,2], the use of one of the propellants as a cooling medium allows to exploit the power taken by the coolant in the cooling system as the input power for gas turbines in expander cycles. Therefore, to ensure the operational life of the engine, the predictions of the flow field and heat transfer in the cooling channels constitute an important part of the design process.

Prediction of pressure loss and heat transfer in cooling channels has been commonly and successfully addressed by one-dimensional models, at least as long as a liquid is flowing in reasonably smooth channels. In LRE, however, when using hydrogen as the coolant, it is typically in the gaseous phase and phenomena like change of properties and thermal stratification must be taken into account [3,4]. Hydrogen has been commonly used, and it is a good coolant, therefore in hydrogen cooled engine design the goal has been mainly devoted to minimize pressure loss through suitably smoothed channels. Channel wall roughness cannot be completely removed, so the effect of a tiny roughness has to be included in modeling.

It is worth noticing that even in the case of tiny roughness, a special model taking into account the different role of roughness on friction and heat transfer has been considered [4–6]. Such a model intends to separate more and more the heat transfer from the friction behavior as roughness increases. It has been shown that, despite the tiny value, roughness modeling can play a significant role in the wall temperature predictions [7]. Roughness, indeed, leads to an increase in friction and heat transfer, although heat transfer enhancement is less than the friction one because of the different physical phenomena [8,9]. In fact, while the friction

* Corresponding author at: Sapienza University of Rome, Italy.

E-mail address: francesco.nasuti@uniroma1.it (F. Nasuti).

Nomenclature

D	Channel diameter	h_s	Equivalent sand grain roughness height
L	Channel length	k_t	Eddy conductivity
Nu	Nusselt number	q	Heat flux
Pr	Prandtl number	u	Velocity
Pr_t	Turbulent Prandtl number	u_τ	Friction velocity
Re	Reynolds number	x, y	Axial, radial coordinates
St	Stanton number		
T	Temperature	<i>Superscripts</i>	
T_τ	Friction temperature	+	Wall normalized values
ΔT^+	Shift of the logarithmic temperature law due to wall roughness	<i>Subscripts</i>	
ΔU^+	Shift of the logarithmic velocity law due to wall roughness	0	Smooth channel values
κ	von Kármán constant	b	Bulk
μ_t	Eddy viscosity	e	Exit
τ	Shear stress	i	Inlet
f_D	Darcy friction factor coefficient	w	Wall
h	Convective heat transfer coefficient	theo	Theoretical or expected values

increase is due to pressure forces acting on the roughness elements, there is no corresponding mechanism with respect to heat transfer. Although the heat transfer capacity of the fluid away from the wall is increased by the turbulence generated by roughness, heat can be transferred by eddy conductivity down to the plane of the roughness elements, but the rate of heat transport in the immediate neighborhood of the surface is controlled by molecular conductivity. Therefore, the Reynolds analogy, or the more general Chilton-Colburn analogy, no longer holds on rough surfaces and the whole concept of analogous behavior between heat and momentum transfer breaks down at high roughness. Although the effects of roughness on momentum transport are well consolidated, literature is less unitary about heat transfer increase. Attempts to directly correlate heat transfer and friction factor coefficients have been reported for over a century [5,8–10] and are the basis for today development [11,12].

The importance of correct modeling of roughness in cooling channels is increasing nowadays, driven by two main different reasons: one is the introduction of additive layer manufacturing techniques which typically come with the realization of high-roughness channels; the other is the use of a coolant less known and more peculiar than hydrogen and water, namely methane.

In fact, on the one hand, metal additive layer manufacturing (ALM) is becoming one of the key technologies for the realization of cooling channels of increasing complexity and efficiency for regeneratively cooled thrust chambers [13,14]. Inherent in ALM processes is the large surface finish, which, when considering minichannels (with hydraulic diameter less than 3 mm), often implies that the absolute and relative roughness are orders of magnitude higher than that considered in conventional designs.

On the other hand, methane has been proposed for different rocket engines and different engine cycles, with all of them using it as the coolant in the regenerative cooling system, followed by injection in the combustion chamber or diversion to a turbine to provide power to an expander cycle engine [15–19]. However, methane has a lower thermal capacity with respect to hydrogen, thus poorer cooling performance, but still high enough to be considered among the candidate fuels for expander cycles [1,2,16,20,19]. It is interesting to underline that, at the cost of high pressure loss, the high roughness can be considered as a design option with a benign effect on the cooling system performance, either for a better cooling of walls or to extract more power to be used in turbine.

Moreover, recent studies on heat transfer deterioration, possibly occurring while heating liquid propellants, have shown that wall roughness has a mitigating effect with respect to heat transfer deterioration, especially at high roughness. In particular, the presence of roughness can lead to a significant reduction of the peak wall temperature and smooths the transcritical process that the coolant may undergo in cooling channels [2,21–23]. Here, it is also worth mentioning that methane may experience a significant change of properties along the channel, as shown by a typical evolution of the Prandtl number depicted in Fig. 1.

In this framework, the numerical predictions of the flow in LRE cooling channels with high roughness become of paramount importance during the design phase. Among the flow models used as a support to the engine design phase, an important role is played by Reynolds Averaged Navier-Stokes (RANS) equation solvers, often resulting in a good compromise between fidelity in reproducing turbulent flow phenomena and computational cost. Originally, RANS approaches do not foresee boundary conditions to take into account for wall roughness, however, different models have been developed in the years mainly focused on a suitable reproduction of wall friction. At present, the most common approaches to account for wall roughness in RANS numerical simulations are based on the *equivalent sand grain approach*. This approach links the real roughness to an idealized equivalent sand grain roughness, with reference to Nikuradse's experiments [24–26]. The roughness effect is mimicked by increasing the turbulent eddy viscosity in the wall region to obtain higher skin friction and heat transfer levels. Although this approach has the advantage of using the same set of equations as for flow over smooth surfaces and the equivalent sand grain roughness heights can be deduced through pressure drop measurements, it preserves the Reynolds analogy with a consequent overestimation of the convective heat transfer increase, especially in the case of high roughness level. Therefore, on this basis, it can only work at low roughness as there is no model to reproduce the difference between skin friction and heat transfer increase at high roughness. Accordingly, a suitable model able to take into account this different behavior must be introduced. This has been done by Aupoix, who developed a modification of Spalart-Allmaras one-equation turbulence model for external flows [27]. Following the same approach, the present study proposes an extension and calibration to internal flows of the high-roughness correction of Spalart-Allmaras model originally developed for external flows in [27]. More specifically, the aim is to improve RANS

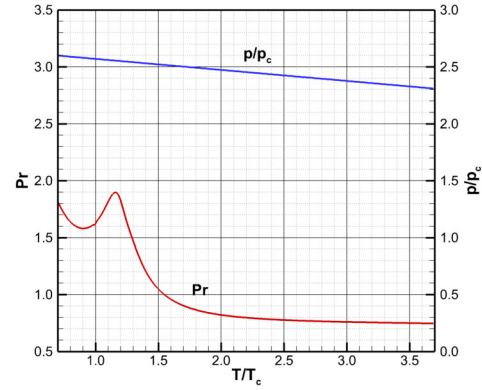
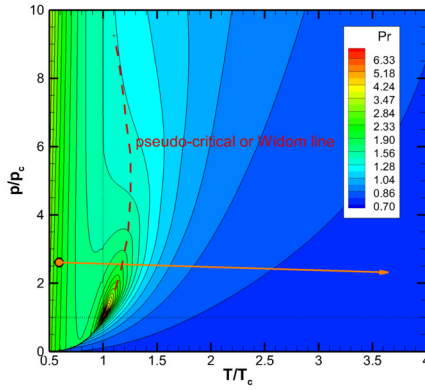


Fig. 1. Possible Prandtl number evolution in LRE cooling channels. (For interpretation of the colors in the figure(s), the reader is referred to the web version of this article.)

model predictions in LRE cooling channels, by developing a thermal correction able to take into account the role of high roughness. Because of the limited amount of experimental data, the proposed model is based on the correlation of Dipprey and Sabersky for heat transfer on fully rough surfaces and validated for circular cross section pipes with constant property flow.

The paper is organized as follows. Wall roughness effect on skin friction and heat transfer and its modeling is reviewed in Section 2. In Section 3 the numerical model and the formulation of the proposed correction are presented. Section 4 is dedicated to the model calibration. The test case employed to perform RANS simulations is described and the grid convergence results with the basic roughness correction are illustrated. Then, the procedure adopted to calibrate the thermal correction and its general expression are described. Finally, results are presented and discussed in Section 5.

2. Wall roughness effects and modeling

Turbulent flows over rough surfaces have been studied since the 19th century, as roughness can be encountered in a wide variety of engineering applications: pipe casting, erosion of gas turbine blades, ice accretion of aircraft, re-entry vehicle nosetips [27–30].

The first main work on roughness, which still constitutes a reference, is due to Nikuradse [24], who investigated a water flow inside a pipe roughened with a calibrated sand grain. Through his experiments, he evidenced three flow regimes according to the sand grain height expressed in wall variables, h_s^+ , also known as roughness Reynolds number:

$$h_s^+ = \frac{u_\tau h_s}{\nu} \quad (1)$$

where h_s is the sand grain height, $u_\tau = \sqrt{\tau_w/\rho}$ the friction velocity, ρ the fluid density, ν the fluid kinematic viscosity and τ_w the wall shear stress. The three flow regimes are known as: hydraulically smooth regime ($h_s^+ < 3.5$); transitional regime ($3.5 < h_s^+ < 68$); and fully rough regime ($h_s^+ > 68$). In particular, in fully rough conditions, the height of the roughness elements definitely exceeds the laminar sublayer thickness and the friction factor f becomes independent of the Reynolds number and is only a function of the relative pipe roughness h_s/D , where D is the pipe diameter. Nikuradse also pointed out that, regarding the velocity profile in terms of the nondimensional wall variables $u^+ = u/u_\tau$ and $y^+ = yu_\tau/\nu$ where y is the distance from the wall, the logarithmic region still exists but is shifted by the quantity ΔU^+ , as shown in Fig. 2.

Near the wall, the viscous sublayer is highly perturbed by the presence of roughness and in the fully rough regime it is entirely

Fig. 2. Dimensionless velocity profile over smooth and rough pipes plotted in wall variables.

destroyed by the large turbulent mixing. This region, strongly affected by roughness elements, is called *roughness sublayer*.

Later, Colebrook performed experiments on rough pipes and correlated the resulting pressure drop data with his well-known Colebrook and White equation [31]. Moody presented Colebrook's results in the *Moody diagram*, which represents the Darcy friction factor f_D as a function of Reynolds number and relative pipe roughness h_s/D , in the range 0–0.05 [32].

Despite a significant number of experimental studies, most of the results reported in the literature mainly focus on roughness effects on momentum transport, while the field of thermal roughness effects is still quite unexplored and not fully understood. Over the years, different correlations of the Stanton number as a function of roughness height, Reynolds and Prandtl number have been derived [5,8,10]. Unfortunately, they all differ in their results, particularly for non-unitary Prandtl number fluids. In 1963, both Dipprey and Sabersky [8] and Owen and Thomson [9] independently proposed the *sublayer Stanton number* concept to account for heat transfer in roughness cavities. Rough walls can be imagined to consist of small cavities whose depth is of the order of the roughness height, in which eddies scour the surface and transport heat between the wall and the vigorous turbulent flow behind the roughness crests, forming the basic convective mechanism of heat transfer at the wall. These eddies are fed with vorticity from the streamwise flow away from the roughness elements and are subjected to frictional resistance at the solid surface where heat transfer between the fluid and the surface occurs by conduction

in the conductive sublayer. Indeed, in the fully rough regime the viscous sublayer is destroyed while the conduction sublayer still exists in the thermal boundary layer. As a consequence, in the proximity of the wall, heat transfer is limited by conduction and dependent on the fluid properties. Kays and Crawford [11] proposed a simplified formulation for the thermal law of the wall, in terms of non-dimensional wall temperature T^+ , to obtain the mean temperature profile over rough surfaces. The wall variable T^+ is defined as $T^+ = (T_w - T)/T_\tau$, where T_w is the wall temperature, T is the local flow temperature, $T_\tau = -q_w/(\rho c_p u_\tau)$ is the friction temperature, q_w the wall heat flux and c_p the fluid heat capacity. This simplified thermal law of the wall reads as:

$$T^+ = \frac{Pr_t}{\kappa} \log \left(\frac{32.6y^+}{h_s^+} \right) + \Delta T_0^+ \quad (2)$$

where Pr_t is the turbulent Prandtl number, κ is the von Kármán constant and ΔT_0^+ , analogous to the momentum wall roughness function ΔU^+ , is a function of the non-dimensional sand grain height h_s^+ and of the fluid Prandtl number Pr .

The mentioned universal friction and thermal laws of the wall constitute a reference for any numerical approach of modeled turbulent flows. Along with them there is today a further reference thanks to the development of direct, i.e. nonmodeled, numerical simulations of turbulent flows (DNS). DNS allows to get high fidelity solutions of the flow around roughness elements for well-identified patterns as well as for randomly generated roughened walls [33–35]. These results are more useful for the verification and calibration of turbulence models than for use in practical applications, especially in the design phase because of their considerable computational cost. As a matter of fact, the correlation developed by Dipprey and Sabersky for fully rough surfaces gives good results with respect to DNS results in terms of Stanton number, further corroborating its use as representative of realistic heat transfer measurements [36,37].

Modeled turbulence relies therefore on the law of the wall or on experimental/DNS data. The most common approach is to introduce roughness models in RANS solvers by the discrete element approach and the equivalent sand grain approaches. The discrete element approach solves equations for a flow spatially averaged over the roughness elements [38,39]. It accounts for roughness by extra terms in the flow equations which represent the flow blockage and the drag and heat flux on roughness elements. However, the alteration of the equations generally precludes its use in general purpose RANS solvers.

Therefore, the most popular and engineering approach is the so called *equivalent sand grain approach*, in which the real roughness height is linked to an idealized one through empirical correlations [25,26] or pressure loss measurements, with respect to Nikuradse's experiments. Then, the equivalent sand grain height is introduced in the turbulence model to mimic the roughness effects by increasing the turbulent eddy viscosity in the wall region to obtain higher skin friction and wall heat flux levels. Unlike the first two approaches, the equivalent sand grain one still preserves the Reynolds analogy. Therefore, it does not reproduce the differences between friction and heat transfer increase, with a consequent overestimation of heat transfer at high roughness [27,40]. In this regard, Aupoix proposed a correction for the turbulent Prandtl number near the wall, based on the introduction of two additional parameters, besides the equivalent sand grain height, to characterize roughness thermal effects [27]. He used the discrete element approach to generate a large database to calibrate the model. However, although Aupoix's correction improves heat transfer prediction, it needs parameters which are not easily accessible unless the roughness pattern is well identified. Aupoix's

correction has been deduced for air ($Pr=0.725$) and mainly validated on external fluxes.

3. Numerical model

The present analysis and model extension has been performed by an in-house preconditioned RANS flow solver, whose validation is reported in [7]. It is based on a Godunov-type finite volume method which is second order accurate in space [41]. Fluid thermodynamic properties are stored in a look-up table, thus the solver is capable to deal with any compressible fluid whose behavior is described by a generic equation of state. Turbulence is computed according to the one-equation model of Spalart-Allmaras [42], equipped with the extension developed by Boeing to account for wall roughness [40]. The latter is corrected to take into account for the thermal effects according to Aupoix's approach for external flows [27]. Parameters however have been recalibrated in the present study to fit experimental laws valid for channels in the fully rough regime. The general expression for the correction is based on the introduction of a local value of the turbulent Prandtl number, which increases in the proximity of the wall, with the purpose of decreasing the eddy conductivity and heat transfer to the wall. The correction of the turbulent Prandtl number takes the form:

$$Pr_t = Pr_{ts} + \Delta Pr_{tr} \quad (3)$$

where Pr_{ts} is the constant value assumed for smooth walls and assumed here as $Pr_{ts} = 0.9$ and ΔPr_{tr} is a correction which must be restricted to the wall region. Hence, ΔPr_{tr} can be expressed as:

$$\Delta Pr_{tr} = F \exp(-y/h_s) \quad (4)$$

where y is the normal distance from the wall. The exponential function aims to reduce the turbulent Prandtl number increase as moving away from the wall while the function F will be calibrated in the following on the basis of the available data. It will be shown in the following sections that F behavior is well described by a parabolic curve expressed as:

$$F(\Delta U^+) = a\Delta U^{+2} + b\Delta U^+ \quad (5)$$

The coefficients a, b have been found to depend on the fluid Prandtl number in the calibration phase, as will be discussed in the following.

Such a correction can be implemented in the model, computing the local value of ΔU^+ at runtime through the relation derived by Grigson [43]:

$$\Delta U^+ = \frac{1}{\kappa} \log \left(1 + \frac{h_s^+}{e^{1.3325}} \right) \quad (6)$$

where the expression of u_τ is in turn a local value which follows from the Boeing extension of the Spalart-Allmaras turbulence model [40]:

$$u_\tau = \frac{\tilde{v}}{\kappa(y + 0.03h_s)} \quad (7)$$

4. Model calibration

To calibrate the model, RANS simulations are carried out for a uniformly heated axisymmetric channel, whose operating conditions are changed to analyze a wide range of roughness levels and Reynolds numbers. The enforced boundary conditions are reported in Fig. 3.

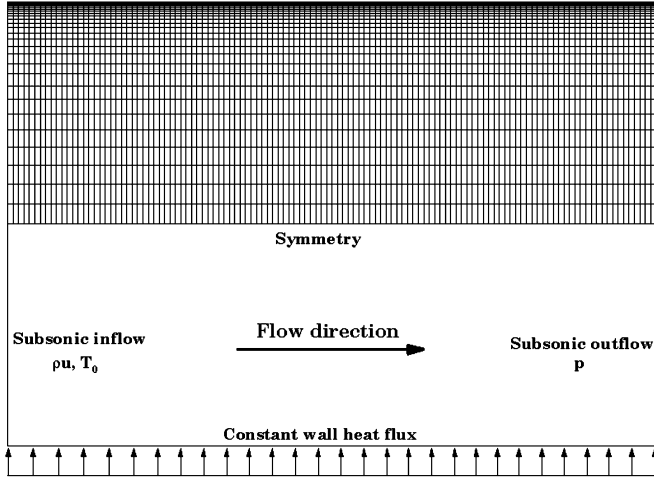


Fig. 3. Grid design and boundary conditions.

At inflow, the static temperature T_i and the mass flow rate per unit area G are enforced, while at outflow the exit pressure p_e is prescribed. The wall boundary condition enforces a constant heat flux q_w , heating the flow. The computational grid is clustered towards the wall to properly solve the boundary layer.

Calibration is made considering a fluid with constant properties ($Pr = \text{const}$) and comparing the numerical results with those of correlations. Before passing to test and calibrate the proposed thermal correction, the grid resolution has been analyzed by a grid convergence analysis using the Spalart-Allmaras model with the original roughness correction proposed by Aupoix and Spalart in [40].

4.1. Grid convergence

The grid refinement study has been performed with the number of cells of the coarse, reference and fine grids which are reported in Table 1.

The resolution of the different grids is obtained while keeping the same clustering law. Therefore, the resulting values of the dimensionless wall distance y^+ change according to the grid level. Grids have been designed as such to get the value of y^+ approximately equal to 1 for the reference grid for the following enforced flow conditions. Grid convergence test case is made assuming $Re = 2.7356 \times 10^4$ (Reynolds number based on the cross

Table 1

Number of cells for the three grid used for RANS simulations in the grid refinement study.

	N_x	N_r
Grid A (coarse)	30	7
Grid B (reference)	60	14
Grid C (fine)	120	28

Table 2

Relative pipe roughness levels h_s/D .

h_s/D	0.04	0.08	0.21
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section diameter), $Pr = 6.033$, $h_s/D = 0.04$ and $L/D = 120$ to ensure a fully developed flow at the exit, and an imposed heat flux. Note that in the present conditions the viscous heating due to friction is negligible with respect to the enforced heat flux.

Results of grid convergence analysis are first presented in terms of the main bulk properties, temperature and pressure. Fig. 4 shows the evolution of the coolant pressure and the bulk temperature along the channel for the three grid levels. As expected, no effect of the mesh on the result of bulk temperature is detected, indicating that the energy balance is correctly satisfied, independently of resolution. Pressure evolution (Fig. 4b) shows a visible displacement at the coarsest level, as can be expected due to its extreme coarseness. However, variations between the reference and the fine grid are quite small, justifying the selection of Grid B as the reference one.

A further analysis of grid convergence has been done on the evaluation of the Darcy friction factor f_D , considering three different relative pipe roughness, as reported in Table 2. Note that the extension of Spalart-Allmaras turbulence model developed by Boeing to account for wall roughness has been calibrated to give good results in terms of skin friction in the fully rough regime, where the Darcy friction factor f_D becomes almost independent of the Reynolds number.

The numerical value of the Darcy-Weisbach friction factor f_D is evaluated from the computed streamwise pressure drop, according to:

$$\frac{dp_0}{dx} = -f_D \frac{\rho u_b^2}{2D} \quad (8)$$

and these results are compared with the Colebrook and White equation for rough pipes [44],

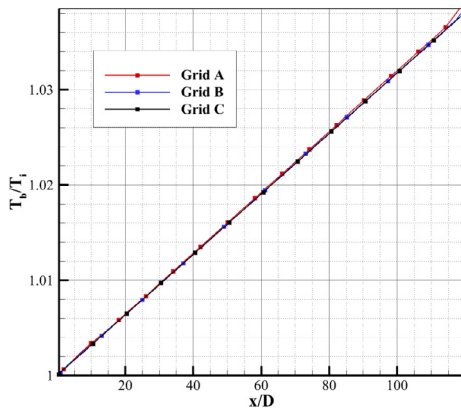
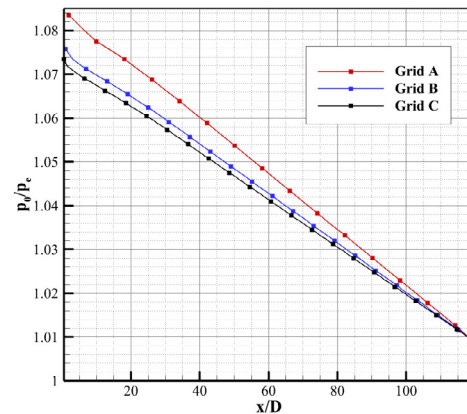
(a) Bulk temperature, T_b/T_i (b) Coolant total pressure, p_0/p_e

Fig. 4. Grid convergence analysis for bulk temperature and pressure evolution along the channel.

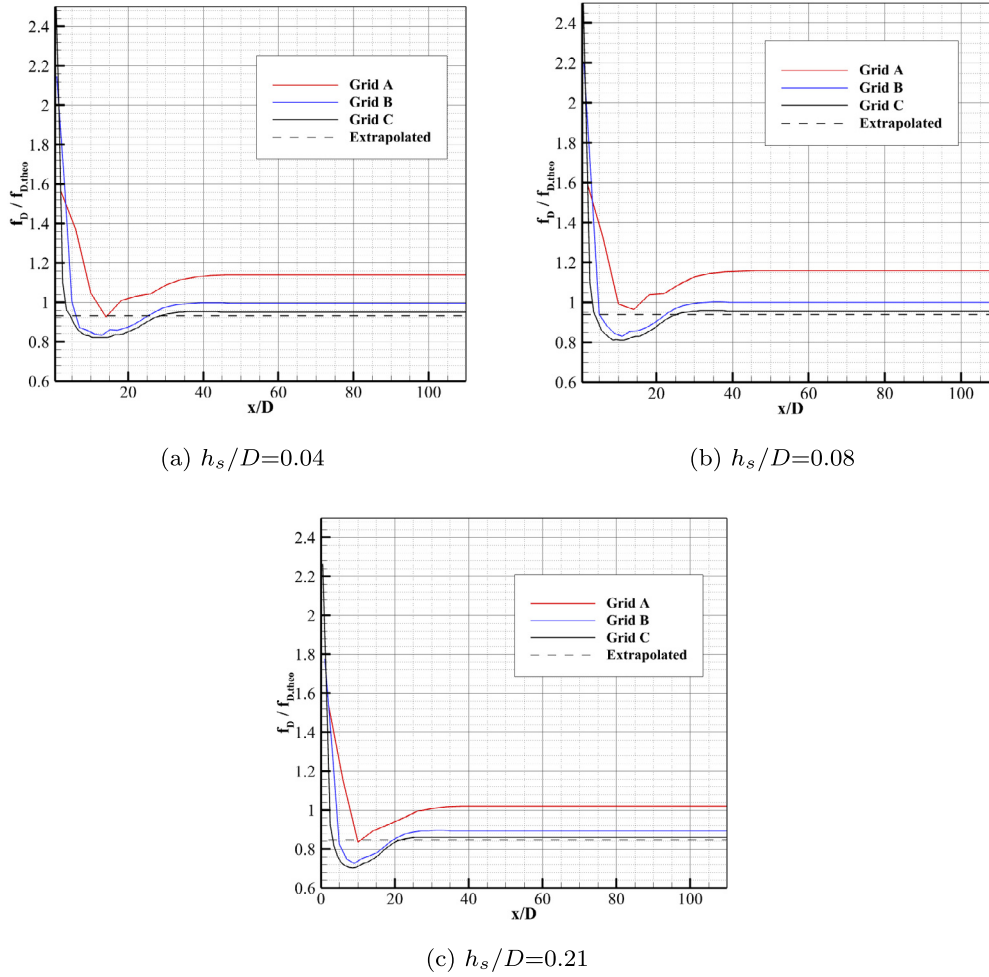


Fig. 5. Grid convergence analysis for the Darcy friction factor.

$$\frac{1}{\sqrt{f_D}} = -2 \log \left(\frac{h_s}{3.71D} + \frac{2.51}{Re \sqrt{f_D}} \right) \quad (9)$$

In Fig. 5 the friction factor is presented as its ratio to the corresponding theoretical (empirical) value obtained by the Colebrook and White equation.

The friction factor behavior is based on the derivative of pressure, therefore it follows the trend of Fig. 4b. Fig. 5 allows to appreciate an almost second-order convergence of the value of f_D , which quickly reaches its asymptotic value. It is to note that the asymptotic value, starting from a value higher than the theoretical one, converges towards an underestimation.

Despite the extension of the Spalart-Allmaras turbulence model to account for wall roughness has been calibrated on external flows, results show a reasonable agreement with Eq. (9) for $h_s/D=0.04$ and 0.08 (about 5% shift). On the other hand, when extremely high values of the relative pipe roughness are considered, the Richardson extrapolated value of f_D is considerably lower than the value of $f_{D,theo}$ obtained solving for f_D the Colebrook and White equation, see Eq. (9). This is due to the fact that the extension of the Spalart-Allmaras turbulence model cannot take into account that in pipe flows the boundary layer growth is constrained by walls and the relative pipe roughness can reach extreme values. In addition, the case of $h_s/D=0.21$ is well outside Colebrook's results reported in the Moody diagram, therefore the reliability of the reference equation could also be questionable. Nevertheless, considered the out-of-range value of roughness and the wide range of applications of the model, the $\sim 16\%$ shift

can still be considered a quite good approximation of the expected friction coefficient.

The grid convergence analysis has been then performed for the Nusselt number $Nu = hD/k$, where the convective heat transfer coefficient has been computed as:

$$h = \frac{q_w}{(T_w - T_{ob})} \quad (10)$$

Fig. 6 shows grid convergence results in terms of the ratio between the numerical computed Nusselt number and the theoretical reference equation of Dipprey and Sabersky [8], here presented in terms of the Stanton number:

$$St = \frac{f_D/8}{1 + \sqrt{(f_D/8)} \left\{ k_f \left[Re(h_s/D) \sqrt{(f_D/8)} \right]^{0.2} (Pr)^{0.44} - 8.48 \right\}} \quad (11)$$

with $k_f=5.19$ according to [8].

The nearly second-order convergence seen for f_D is similarly appreciable for Nu . Differently, the asymptotic value of Nu is extremely far from the theoretical one and the error increases with increasing equivalent sand grain roughness height h_s , reaching a value greater than 400% for $h_s/D=0.21$. This is a main drawback of the equivalent sand grain roughness approach: if the equivalent sand grain roughness height is adjusted for skin friction, the predicted heat transfer results to be too high. Indeed, keeping the

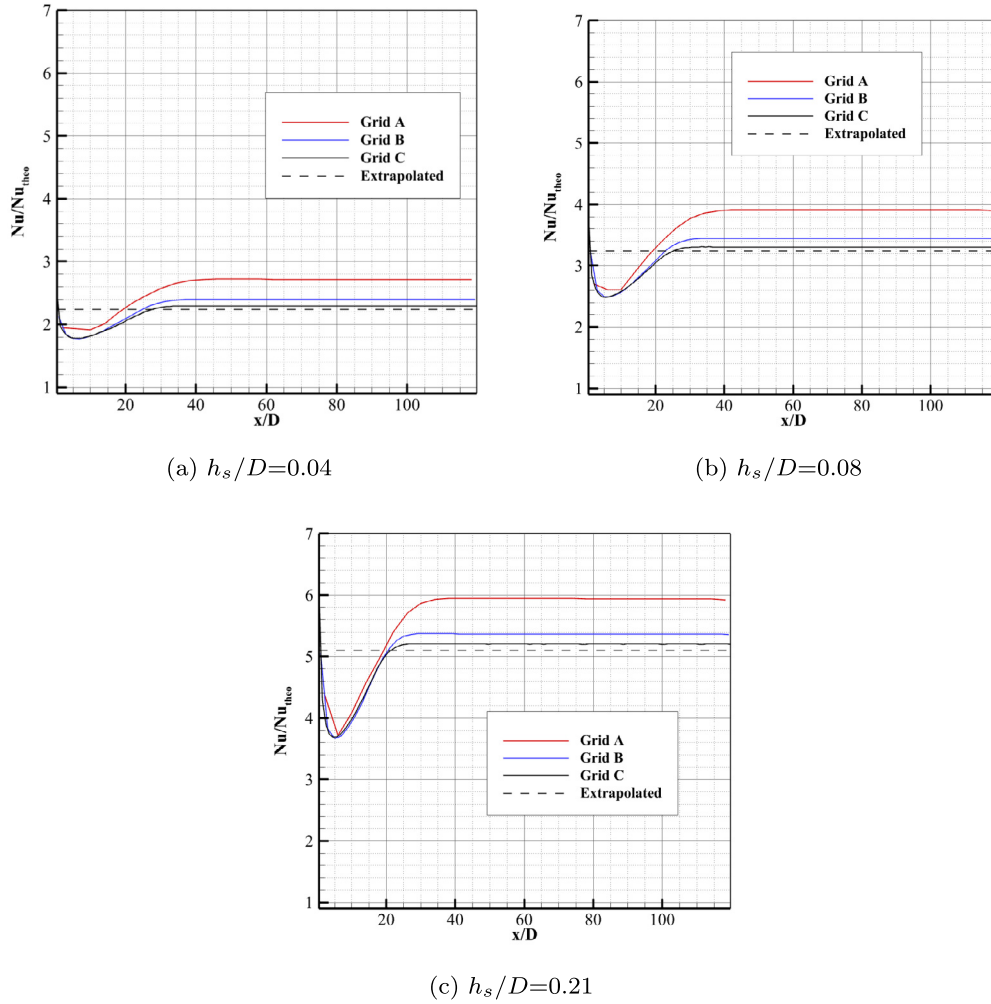
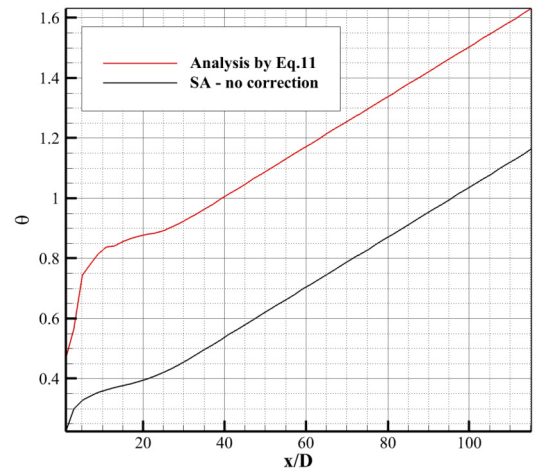


Fig. 6. Grid convergence analysis for the Nusselt number.

turbulent Prandtl number Pr_t constant and equal to 0.9, the turbulent conductivity k_t is directly derived once the turbulent viscosity μ_t is known. As a consequence, the different effects of roughness on skin friction and heat transfer are not reproduced and heat transfer results to be considerably overestimated at high roughness. Additionally, the error on heat transfer prediction dramatically increases with increasing equivalent sand grain roughness height. As a matter of fact, experiments have shown that there is a limit for any combination of Reynolds and Prandtl numbers beyond which an increase in roughness, while increasing the friction factor f_D will no longer increase the heat transfer coefficient h [8]. Indeed, when the roughness effect on friction becomes very large ($f_D/f_{D,0} > 4$, with $f_{D,0}$ the Darcy friction coefficient for smooth channels), no further increase in heat transfer is observed because the heat transfer resistance has become primarily a conduction resistance at the surface in the spaces between the roughness elements [11].

Such an overestimation of the heat transfer at the wall leads to underestimated wall temperature, as shown in Fig. 7, where the analysis with Eq. (11) has been made considering the bulk quantities obtained from the CFD simulation. Comparison has been made through the variable $\theta = (T_w - T_{b,i})/(T_{b,e} - T_{b,i})$, for $Re = 2.7356 \times 10^4$, $h_s/D = 0.08$. This aspect is particularly critical in the design of a liquid rocket engine because an optimistic prediction of the fluid heat transfer capacity would lead to a wrong prediction of the wall temperature distribution with the risk of thermal failure.

Fig. 7. Numerical and theoretical results of the non-dimensional wall temperature $\theta = (T_w - T_{b,i})/(T_{b,e} - T_{b,i})$, $Re=2.7356 \times 10^4$, $h_s/D=0.08$.

4.2. Thermal model calibration

To improve heat transfer prediction, the approach developed by Aupoix has to be recalibrated to fit the correlation of Dipprey and Sabersky in the fully rough regime. As mentioned in Section 2, Aupoix's correction brings significant improvements in the heat

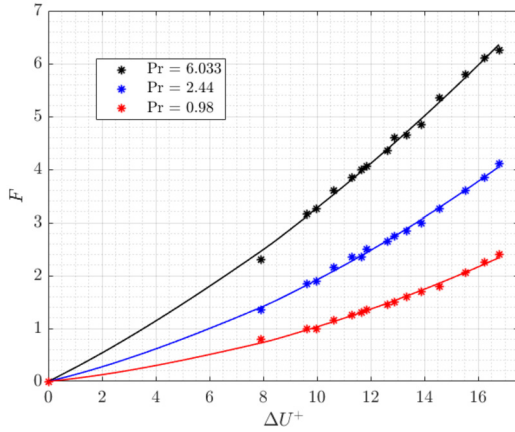


Fig. 8. Fit of the F function for different Prandtl number.

transfer predictions, however the coefficients are not optimized with respect to channel experimental data as validation has been mainly done on external flows. Additionally, he considered only air at moderate temperature, so $Pr=0.725$. As illustrated in Section 3, the correction to the turbulent Prandtl number is expressed by Eq. (4), where the function F must be now numerically determined to get the correlation results. To this goal, for each different combination of h_s and Re considered, the value of F is artificially varied in the numerical model until the convective heat transfer coefficient in the fully developed region matches the theoretical value of Dipprey and Sabersky's correlation. Once the values of F have been determined, they have been correlated with the corresponding values of the velocity shift ΔU^+ , to characterize roughness dynamical effects. As the heat transfer is strongly dependent on the fluid Prandtl number, the correction has firstly been developed considering $Pr = 6.033$, representative of water. The effect of the Prandtl number on the correction has been then investigated considering $Pr = 2.44$ and $Pr = 0.98$, representative of the possible methane Prandtl number in LRE cooling channels (see Fig. 1).

Fig. 8 shows the numerical values of F for the three different values of Prandtl number considered, as linked to the velocity shift ΔU^+ . Data are fitted to characterize the curve, including the point $(\Delta U^+, F) = (0,0)$ as the correction must approach zero for smooth surfaces.

It results that a good fitting is obtained by a parabolic curve, as anticipated in Eq. (5), with coefficient a , b which are functions of Pr :

$$a = (-2.346 \times 10^{-4})Pr^2 + (2.102 \times 10^{-3})Pr + 3.542 \times 10^{-3}$$

$$b = -(2.303 \times 10^{-3})Pr^2 + (5.588 \times 10^{-2})Pr - 3.043 \times 10^{-3}$$

Fig. 9 shows the comparison between numerical results obtained with the basic roughness correction model (SA – no correction) [40] and the thermal correction for high roughness channels proposed in this work (SA – Pr_t correction), in terms of the Nusselt number Nu , for $Re = 8.207 \times 10^4$. The convective heat transfer coefficient is considerably reduced and in line with the theoretical values of Dipprey and Sabersky correlation.

5. Discussion

In this section the numerical results obtained using the present thermal correction for high roughness channels (THRC) with Spalart-Allmaras turbulence model are discussed. The effect of roughness is analyzed in terms of dimensionless velocity and temperature profiles, then the friction factor and Nusselt number values are shown compared to the theoretical values of a smooth pipe. Results obtained with THRC correction are also presented for:

1) comparison with correlations and the basic Aupoix approach at Pr number inside and outside of the calibration range; and 2) a test case representative of a liquid rocket engine cooling channel where methane is considered as a real fluid with variable properties.

Figs. 10, 11 show the solutions obtained for the dimensionless velocity and temperature profiles ($Pr = 6.03$), respectively. The behavior of the dimensionless velocity profile at different roughness levels (Fig. 10) shows that the effect of roughness is well mimicked using the Boeing extension of the Spalart-Allmaras turbulence model [40]. Indeed, roughness has a major effect right at the wall, where the viscous sub-layer ($0 < y^+ < 5$) is physically destabilized by the presence of the roughness elements. In particular, in the fully rough regime, the viscous sub-layer almost disappears entirely, accounting for the fact that the influence of viscosity on the hydrodynamic behavior of the flow is negligible. Above it, the logarithmic region is shifted downwards by the quantity $\Delta U^+(h_s^+)$, but it retains its original slope [24].

The effect of THRC correction is clearly visible from the profiles of the dimensionless temperature T^+ obtained for $Pr=6.03$ with the assumption of smooth and rough pipe, respectively (Fig. 11). When the turbulent THRC correction is not applied, and therefore the turbulent Prandtl number is kept constant and equal to 0.9, the temperature shift ΔT^+ with respect to the smooth curve is overestimated and follows the velocity profile. On the other hand, when the THRC correction is applied, it yields to a decreased heat transfer toward the wall, and, accordingly, the increase in the wall temperature T_w results in a smaller temperature shift ΔT^+ .

Fig. 12 shows the increase in friction factor f_D caused by roughness obtained for a flow with $Re = 8.207 \times 10^4$. The numerical results of the reference grid are compared with the theoretical ones obtained using Eq. (9). Both theoretical and numerical values are normalized with respect to the theoretical value for the corresponding smooth tube, ($h_s/D=0$). Numerical results have been evaluated in the fully developed region where the friction factor is constant, being in the fully rough regime.

The friction factor increases sharply with increasing equivalent sand grain roughness height and directly influences the pressure loss. As mentioned in Section 4, numerical results are in good agreement with the theoretical ones until extremely high values of the relative pipe roughness.

Results obtained with the THRC correction are shown in Fig. 13 in terms of the Nusselt number ratio, also normalized with respect to the theoretical value for smooth pipes Nu/Nu_0 , as a function of the relative pipe roughness h_s/D . The reference value Nu_0 is computed by the Dittus-Boelter equation, while numerical results have been computed in the fully developed region. According to the experimental results of [45], the impact of roughness on heat transfer varies with Prandtl number, due to the corresponding variation of the thickness of the heat conduction sublayer. Additionally, the effect of roughness on heat transfer performance tends to flatten out for high values of the equivalent sand grain roughness height [11,46]. The friction factor ratio $f_D/f_{D,0}$ discussed in Fig. 12 is also reported here for comparison. The comparison highlights the different behavior of friction factor and Nusselt number increase with increasing roughness.

This comparison is made more evident in Fig. 14 showing the ratio between heat transfer and friction factor increase caused by roughness normalized with respect to the smooth channel. Flattening of the heat transfer increase and continuous increase of friction factor lead to the decrease of their ratio at high values of the relative pipe ratio roughness. It is worth noticing again that the effect of roughness is reduced for lower values of the Prandtl number.

The results obtained with THRC are then compared to the original thermal correction presented by Aupoix in [27] for one of the calibration test cases. In particular, the test case consid-

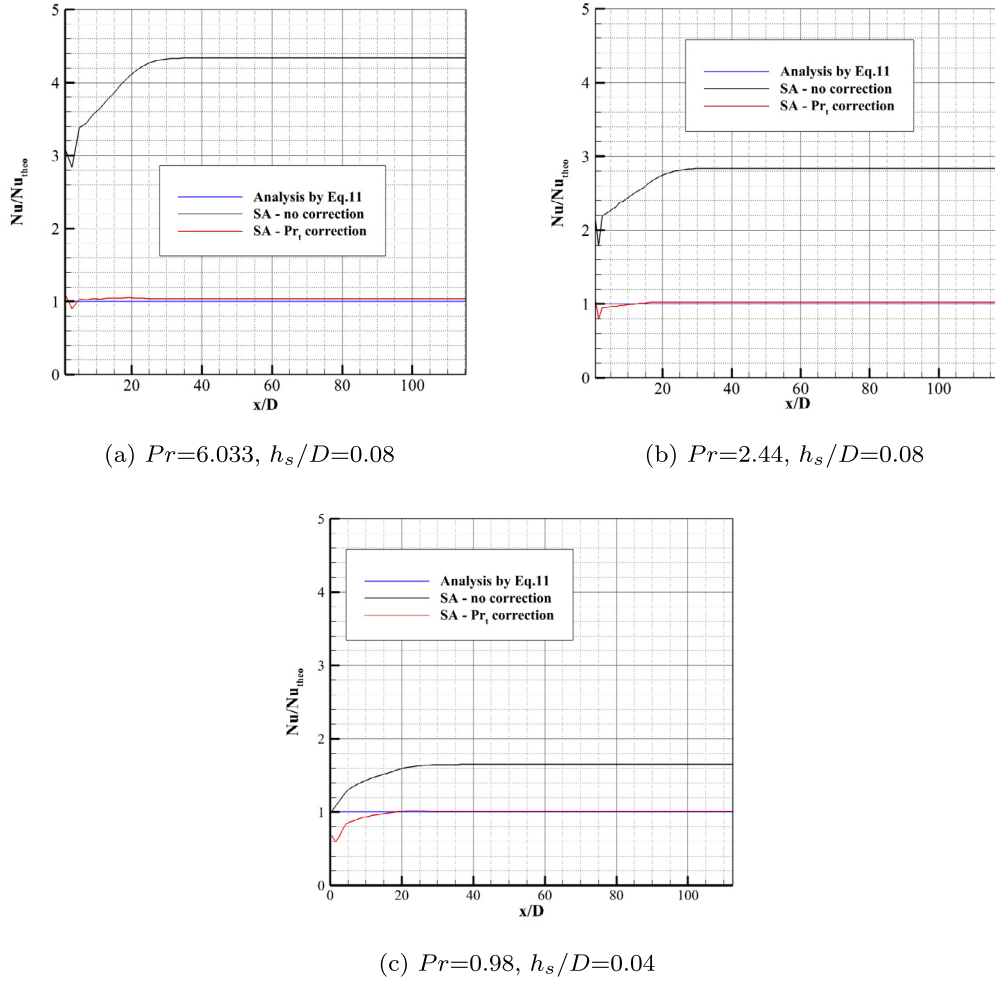


Fig. 9. Comparison of the convective heat transfer coefficient h computed through the original and corrected model, $Re = 8.207 \times 10^4$.

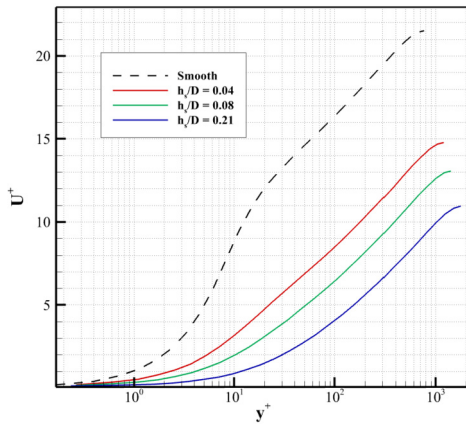


Fig. 10. Smooth and rough dimensionless velocity profile U^+ as a function of the dimensionless wall normal coordinate y^+ , $Re=2.7356 \times 10^4$.

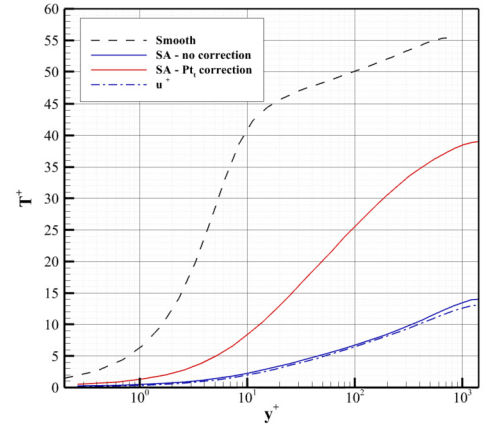


Fig. 11. Smooth and rough dimensionless temperature profile T^+ as a function of the dimensionless wall normal coordinate y^+ , $Re=2.7356 \times 10^4$, $h_s/D=0.08$, $Pr=6.03$.

ers $Re=2.7356 \times 10^4$, $h_s/D=0.08$, $Pr=6.03$. For this Prandtl number, which is within the calibration range and quite different from that used for external flows by Aupoix, it can be seen in Fig. 15 that also considering the whole range of the parameter s_{corr} , introduced to represent the kind of structured roughness under scrutiny, the computed value of the Nusselt number remains always far from the expected value according to Dipprey and Sabersky correlation, Eq. (11).

Moving to a test case within the calibration range of Aupoix reported in [27], $Re=8.207 \times 10^4$, $h_s/D=0.08$, $Pr=0.725$, results obtained with the present THRC correction fall within the span of results obtained with the Aupoix's correction, see Fig. 16. Additionally, THRC numerical results are also in line with Nunner's correlation [5], which has been derived for pipes and a fluid Prandtl number equal to 0.7.

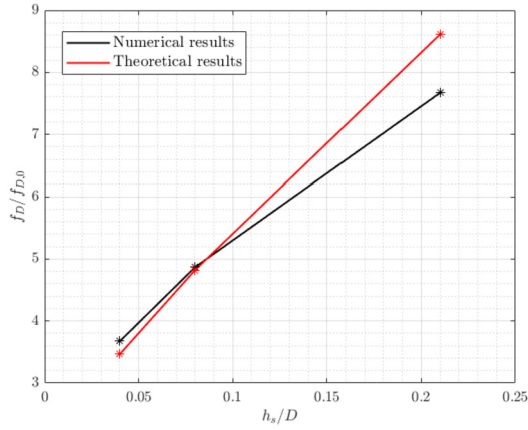


Fig. 12. Friction factor ratio $f_D/f_{D,0}$ for the three different values of h_s .

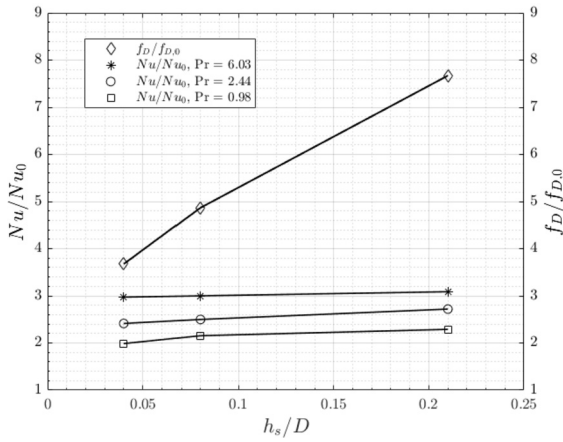


Fig. 13. Nusselt number ratios for different values of h_s , Pr , $Re=8.207 \times 10^4$.

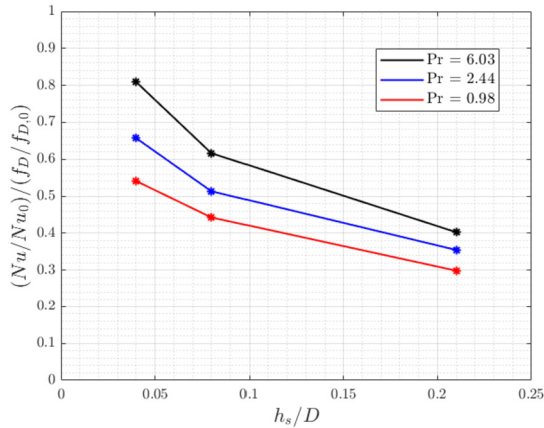


Fig. 14. Nusselt number ratios for different values of h_s , Pr , $Re=8.207 \times 10^4$.

During the study and the calibration of the correction, different simplifying assumptions have been made: the fluid properties are assumed to be constant and a constant uniform heat flux is enforced as a boundary condition. To test the correction outside of the simplifying hypothesis of constant fluid properties, simulations considering super-critical methane with variable properties have been carried out. Specifically, the 32-term modified Benedict-Webb-Rubin (mBWR) equation of state [47,48] has been used.

The considered test case is characterized by $h_s/D=0.063$, $L/D=250$ and by an inlet velocity of 8.3 m/s. A representative exit pressure of 150 bar has been assigned for the exit boundary

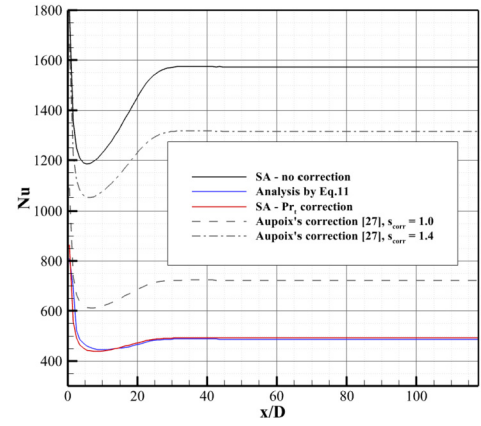


Fig. 15. Nusselt number, $Re=2.7356 \times 10^4$, $h_s/D=0.08$, $Pr=6.03$: comparison between the SA original model [40], present Pr_t correction (THRC), Aupoix correction [27].

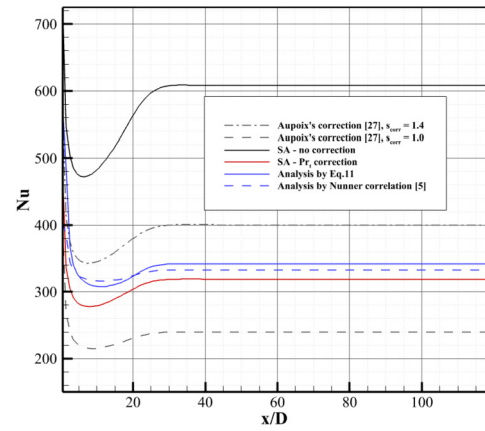


Fig. 16. Nusselt number, $Re=8.207 \times 10^4$, $h_s/D=0.08$, $Pr=0.725$: comparison between the SA original model [40], present Pr_t correction (THRC), Aupoix correction [27], Nunner correlation [5].

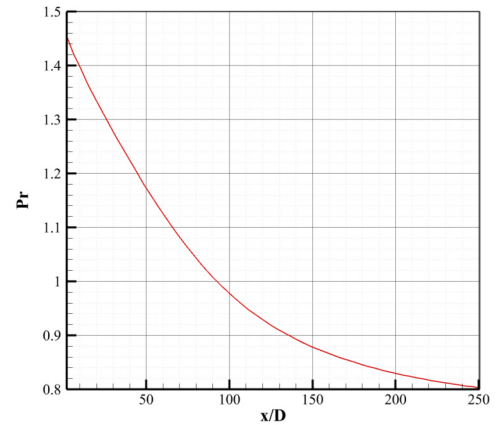


Fig. 17. Variable properties super-critical methane: Prandtl number evolution along the channel.

condition. The inlet temperature is 250 K and the Reynolds number reaches values up to 220000. Due to the variation of methane properties, the Prandtl number decreases from 1.45 to about 0.8, as shown in Fig. 17. A constant heat flux of 1 MW/m² is imposed at the wall. Results are discussed in terms of the convective heat transfer coefficient and the Nusselt number, the latter reported in Fig. 18. The numerical results are compared with those obtained with the correlation of Dipprey and Sabersky. The correlation is used with bulk properties resulting from the numerical simulation.

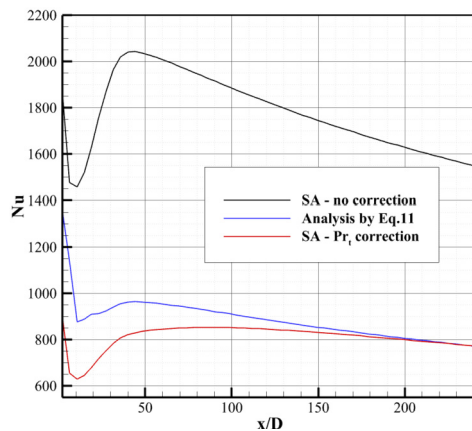


Fig. 18. Variable properties super-critical methane: comparison of the Nusselt number along the channel computed through the original and corrected model.

Results are interesting as they show the ability of THRC correction to handle properly the changes of properties. It is also interesting to discuss the discrepancy with respect to the Dipprey and Sabersky correlation that occurs in the first part of the channel. At a first glance, it can be surprising as the THRC correction has been calibrated on that correlation. Indeed, this discrepancy should be expected. The results are in good agreement in the region where the changes of the Prandtl number in streamwise and crosswise direction are smaller, while in the first part of the channel the Prandtl number changes are larger. In a region of large sensitivity of Prandtl number changes with temperature, or in correspondence of high heat fluxes leading to large changes of temperature in the crosswise direction, the hypothesis and data under which the Dipprey and Sabersky correlation is developed are no longer valid. In principle, the effect of roughness is mainly driven by the phenomena occurring in the vicinity of the wall, so local rather than bulk properties should be used. Accordingly, THRC correction takes into account for the local properties.

In general, however, the THRC correction gives results in line with expectations, showing a good agreement in a regime of constant properties. Such a result allows to consider the present correction as a general value correction for the application in circular cross section cooling channels with high roughness. On the other hand, in case of changing properties, the availability of suitable experimental or DNS data would be of support to confirm that the predictions of THRC corrected rough model of Spalart-Allmaras RANS closure are more robust than their calibration source.

6. Conclusions

Following the approach proposed by Aupoix [27], a thermal correction based on the correlation of Dipprey and Sabersky has been calibrated and implemented to predict realistic values of the convective heat transfer coefficient in high roughness cooling channels. The correction is based on the turbulent Prandtl number increase in the wall region, which allows to decrease the wall heat transfer. The latter would be highly overestimated at high roughness with the basic extension of equivalent sand grain approach to account for wall roughness. At present, the limited amount of experimental or Direct Numerical Simulation data only allowed to validate the correction on data for circular cross section pipe with a constant property flow. However, it can be expected that the approach will improve the prediction for rectangular cross section channels and variable property flows as well. More in general, the prediction by simulations addressing the actual complex geometries of LRE cooling channels, typically featuring high aspect ratio, curved rectangular channels, asymmetrical heating conditions can

be improved by the present model. Designers can largely benefit from this approach which can be simply implemented in the Spalart-Allmaras model, which is a basic and widespread closure of Reynolds Averaged Navier-Stokes equations. It will allow to reduce the risk of overestimation of cooling capabilities and at the same time take into account phenomena like thermal stratification that cannot easily be considered in one-dimensional approaches.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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